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A Machine Learning Method to Improve Non-Contact Heart Rate Monitoring Using an RGB Camera

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ABSTRACT Recording and monitoring vital signs is an essential part of home-based healthcare. Using contact sensors to record physiological signals can cause discomfort to patients, especially after prolonged use. Hence, remote physiological measurement approaches have attracted considerable attention, as they do not require physical contact with the patient's skin. Several studies proposed techniques to measure heart rate (HR) and heart rate variability by detecting the blood volume pulse from human facial video recordings while the subject is in a resting condition. In this paper, we focus on the measurement of HR. We adopt an algorithm that uses the independent component analysis (ICA) to separate the source (physiological) signal from noise in the RGB channels of a facial video. We generalize existing methods to support subject movement during video recording. Furthermore, we improve the accuracy of existing methods by implementing a light equalization scheme to reduce the effect of shadows and unequal facial light on the HR estimation, a machine learning method to select the most accurate channel outputted by the ICA module, and a regression technique to adjust the initial HR estimate. With respect to the ECG measurement ground truth, the proposed method decreases the RMSE by 27% compared with the state of the art in the stationary condition. When the subject is in motion, our proposed method achieves an RMSE of 1.12 bpm.

INDEX TERMS Blood volume pulse (BVP), independent component analysis (ICA), photoplethysmography (PPG), remote heart rate measurement.

I. INTRODUCTION

Heart rate (HR), defined as the number of heartbeats per unit of time, is a significant measure for assessing cardiac well-being [1]. Typically, it is measured as the number of heart contractions per minute (bpm). Its regular measurement is crucial in assessing and monitoring a myriad of health conditions. Contactless techniques have a variety of applications. For example, in long-term HR monitoring [2], contact-based techniques such as ECG may lead to skin irritations as the electrodes are replaced daily over roughly the same area. Hence, contactless techniques are well suited to this scenario. Also, there are situations where the patient might not be interested in keeping skin irritating sensors attached. Prisoners on suicide watch who are monitored to ensure that they do not self-harm fall under this category. A growing body of literature investigates the possibility of using camera-based contactless techniques, also known as camera-based non-contact photoplethysmography (PPG), to measure HR. Several attempts have been made to increase the reliability

and accuracy of these measurement techniques. These methods are based on the principle that the light absorption properties of blood differ from the surrounding skin. Hence, when the arterioles' blood volume increases, the skin color varies. These changes can be recorded by a standard RGB video camera.

Poh *et al.* [3] use Independent Component Analysis (ICA) [4] as a blind source separation technique to extract the channel containing BVP information from a linear mixture of the underlying signals. Monkaresi *et al.* [5] use the same approach under different lighting conditions and limited user motion. The output of the ICA process is three independent signals. Any of them can carry PPG information. Monkaresi *et al.* [5] apply a Machine Learning (ML) technique to assess which component in estimating HR. They conclude that the third component is the most useful for HR estimation.

Previous video-based HR measurement studies suffer from several limitations. First, the subjects were permitted only

slight movements [3]–[6]. Second, the measured PPG signal was vulnerable to noise caused by lighting condition changes [7]. Third, since the implemented ICA module produces three signals as output, mostly unproven heuristic methods were used to select the signal best reflecting the BVP information [3].

In the proposed approach, we alleviate the assumption of subject stillness. Hence, we suppose that the subject is given the freedom to roam the environment. The algorithm shall measure the HR given that it receives a video that captures at least 50% of the subject's face. We summarize our contributions as follows:

- We propose a light equalization scheme to minimize the negative effect of noise generated from the fluctuation of light on HR estimation;
- We develop an ML algorithm to select the PPG information carrying component after performing ICA on the RGB channels; and
- We propose a linear regression model to improve the accuracy of the HR estimation obtained through the ICA.

The rest of the paper is organized as follows: Section II presents the background and related work; Section III describes the proposed method; Section IV details the dataset used in the evaluation of the proposed method; Section V provides a detailed description of the proposed method's evaluation; Section VI discusses the obtained results; and Section VII concludes the paper with a summary of the results.

II. BACKGROUND AND RELATED WORK

To the best of our knowledge, among the various attempts towards measuring human vital signs remotely, Chen *et al.* [8] were the first to introduce remote measurement of HR. They use a Doppler radar as an active sensor in a microwave system [9], [10]. The radar detects heart and breathing rates from a 30 meters distance [11], [12]. The radar projects a low-intensity microwave signal. The signal is reflected onto the subject and received back by the radar. By analyzing the reflected beam, they measure the motion of the chest wall and abdomen, and consequently, the breathing rate of the subject [13]. Sun *et al.* [14] demonstrate the first remote passive cardiac pulse measurement through a thermal imaging system. This method is based on the computation of the frequency of pulses derived from the temperature changes on the vessel modulated by pulsatile blood flow. However, this approach has some limitations. It requires the use of an expensive thermal imaging camera. In addition, it can yield inaccurate results due to the variation in room temperature [15].

As opposed to the above-described methods, video-based methods measure variations in the subject's skin color to extract a PPG signal. PPG is a non-invasive technique to sense the BVP through the light reflected from the skin. As blood absorbs more light compared to the surrounding tissue, changes in the blood volume affect the reflectance of the light [3]. Takano and Ohta [16] measure HR using a digital

camera in an environment with ambient light. They calculate the average brightness of time-lapse skin images recorded by a CCD camera. They then apply auto-regressive spectral analysis on the images to calculate the HR.

Sun *et al.* [17] investigate the feasibility of remote monitoring of HRV and other physiological signals. They use palm images to extract the HRV. They employ a CMOS camera-based imaging system to capture the PPG signal through palm images.

McDuff *et al.* [18] utilize a five-band camera (Red, Orange, Green, Cyan, and Blue) instead of an RGB camera to measure HR and HRV from facial images. They use the Local Evidence Aggregated Regression (LEAR) facial landmark detection method [19] to find the coordinates of the face landmarks in each frame. They spatially average the color channels and present them to the ICA module. Their results indicate that the combination of Cyan, Green, and Orange colors yields measurements that are more accurate. However, typically, five-band cameras are more expensive than RGB cameras.

Kumar *et al.* [20] extract the HRV using the Maximum Ratio Combining (MRC) method. This method combines the average pixel intensity of signals from different regions of the face. They achieved -5.32 dB overall Signal to Noise Ratio (SNR) for the PPG signal and 8.09 ± 22.44 bpm RMSE.

Antink *et al.* [21] employ a multimodal technique where they extract skin color changes and head movement from video, and record a ballistocardiographic signal from an instrumented chair. They fuse the collected signals using a Bayesian approach to estimate the HR. They obtain an average absolute error below 25 ms with a coverage of 90% compared to the ECG ground truth.

Rubinstein *et al.* [22] propose the Eulerian Video Magnification (EVM) method to reveal and magnify small changes in skin color due to the heart cycle. EVM uses localized spatial pooling and temporal filtering to detect and magnify subtle changes in the skin color that reflect BVP.

Alghoul *et al.* [23] investigate two methods to extract HR and HRV from a facial video. They compare EVM and ICA approaches. Their results show that ICA-based methods are generally more accurate than EVM-based techniques.

Poh *et al.* [24] proposed a method to recover the HR in non-laboratory conditions. They use a face tracker algorithm to track the small motions of the subject during measurement [3]. However, the subject remains in a fixed position relative to the camera. The subject's movements are restricted to tilting the head sideways, nodding, looking up/down, and leaning forward/backward.

Park *et al.* [25] present a non-contact measurement method to monitor cardiac responses using infrared images of the patient's pupil. They analyze the harmonic frequency of the pupillary rhythm to find the response of the heart.

III. METHOD

We base our approach on the system proposed by Poh *et al.* [3]. Fig. 1a depicts the steps necessary to produce an

HR estimation from video using the Poh *et al.* [3] technique. First, they find the face of the still subject using a face detection mechanism. Then, they select a rectangular area of the face, which includes 60% of the width and the full height of the detected face as a Region of Interest (ROI). They decompose the ROI into the RGB channels by averaging the Red, Green, and Blue amplitude values over all the pixels for each frame. They present the RGB signals to the ICA module to separate them into three independent components. One of the independent components produced by the ICA module is most reflective of PPG. However, it is not always clear which one. Poh *et al.* [3] heuristically choose the second component. Conversely, in another study by the same authors, they choose the signal with the maximum integrated power spectrum [24].

Fig. 1b depicts the steps involved in processing a facial video to estimate HR using the proposed approach. To describe our method, in the following subsections, we discuss only the differences between the proposed approach and Poh *et al.* [3]. We highlight our contributions in red in Fig. 1b.

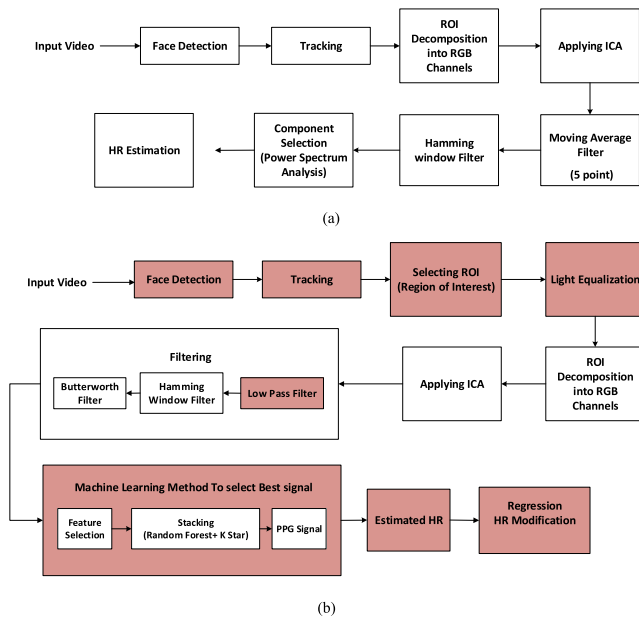


FIGURE 1. HR prediction algorithm for the (a) Poh *et al.* [3] method and (b) proposed method.

A. FACE DETECTION AND TRACKING

We use a face detection method and a face tracker to find and track the face in a video recorded by a standard RGB camera. In our experiment, we use a 24-bit RGB camera recording at a 30-frame per second and 640×480 pixels per frame. The face detection method (OpenFace) [26] finds the face in the first frame. Then, we use the Kanade–Lucas–Tomasi (KLT) [27]–[29] scheme to track the face across the video frames as the subject moves. We elect to consider the entire face of the subject as the ROI to collect enough facial information when the size of the face decreases due to the increase in the distance from the camera.

B. LIGHT EQUALIZATION

Objects look differently under varying light conditions. Human beings recognize objects in images with little effort despite lighting changes. However, it is challenging for computer vision systems to achieve the same feat.

The noise generated by fluctuations in the light levels can disrupt continuous measurement of HR. This is especially important given that the proposed approach, as opposed to previous ones [3], [5], supports subject motion and roaming. Hence, changes in lighting levels on the face are bound to occur in an unevenly lit environment. To address this issue, we decompose the ROI pixels into the H (Hue), S (Saturation), and L (Lightness) color components. We calculate L' by normalizing the values of L in the ROI across the frames of the video segment. We then replace L_{xyi} by L'_{xyi} for every pixel at (x, y) in frame i . We leave the values of the H and S components unchanged.

For a given video segment composed of n frames, we calculate L'_{xyi} using equation (1) where we denote the mean of L_{xyi} for pixel (x, y) over n frames by M_{xy} and the standard deviation by S_{xy} .

$$L'_{xyi} = \frac{L_{xyi} - M_{xy}}{S_{xy}} \quad (1)$$

In a continuous monitoring scenario, the video segment refers to the length of video we use to calculate one HR value. Although different segment lengths can be employed by the system's operator, for our experiments, we adopted a standard segment length of 60 seconds (i.e. approximately 1,800 frames for a 30 frames-per-second (fps) sampling rate). We reconstruct the ROI using the H, S, and L' color components. Therefore, we decrease light fluctuation in the frames of a single video segment. Fig. 2 shows the changes in lighting across three frames before (Fig. 2a) and after (Fig. 2b) light equalization. In Fig. 2a, the subject's face is unevenly lit in all frames (one side is darker than the other). However, that effect is visibly reduced in Fig. 2b. Furthermore, the lighting changes across the three frames are more pronounced in Fig. 2a compared to Fig. 2b.

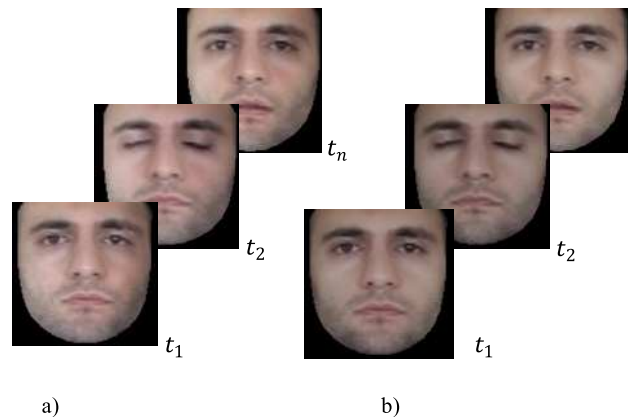


FIGURE 2. Face images before and after light equalization. a) Before light equalization. b) After light equalization.

C. FILTERING

To form the raw signals, the ROI is separated into the RGB channels then spatially averaged over all pixels. The raw signals from the Red, Green, and Blue channels are de-trended using a technique based on the smoothness priors approach [30]. The signals are de-trended by a smoothness parameter of $\lambda = 2000$. Moreover, the de-trended signals are normalized by subtracting the mean and dividing by the standard deviation. After normalizing and de-trending, the RGB signals are presented to the ICA module. The order of the independent components produced by the ICA module is random [36]. It is difficult to measure HR immediately post-ICA, as the produced signals require noise reduction procedures.

We apply a Low Pass Filter with a cut-off frequency of 5 Hz to the ICA output signals. The cut-off frequency is selected based on the frequency range of a typical HR signal. Furthermore, we apply a Hamming window filter to reduce the ripples in the signal and retrieve a more accurate representation of the original signal's frequency spectrum.

Next, we use a narrow band second order Butterworth filter proposed by [23] to further clean up the signal. This filter uses an initial HR estimate to build a narrow band IIR Butterworth filter with the following cut-off frequencies:

$$\begin{aligned} L - \text{Cutoff} &= \alpha \times \text{HR_Estimate} \\ H - \text{Cutoff} &= \beta \times \text{HR_Estimate} \end{aligned}$$

where $\alpha < 1$ and $\beta > 1$.

Per Alghoul *et al.* [23], we adopt an α value of 0.8 and β value of 1.2. The filter continuously updates its *HR-Estimate* (where *HR-Estimate* refers to the HR estimate in the previous video segment) to ensure adaptability to changes in the HR.

D. CHANNEL SELECTION

There are commonly two ways to calculate the HR after filtering is concluded [31]. The first one is to count the number of peaks in the time domain PPG signal. The second one, used by Poh *et al.* [24] and referred to as the MPA (Maximum Peak among All) approach, is to identify the highest amplitude of the frequency domain representation of the PPG. The component with the highest peak in the frequency domain corresponds to the HR. We adopt the latter technique as it is less sensitive to minor disturbances in the time domain signal that might introduce noise artifacts that mimic PPG peaks.

ICA produces three independent components (i.e. signals) in a random order [36]. One of these independent components best reflects the PPG information. However, due to the randomness of the order of the ICA output, it is unclear how to identify that component. Therefore, we propose an ML-based algorithm for independent component selection post-ICA. The algorithm uses the time and frequency domain characteristics of the signals to predict the PPG channel. We heuristically choose ten features to extract from each independent component outputted by the ICA module to

feed to the ML algorithm. The features are categorized into two sets: statistical descriptor features (e.g. mean and standard deviation of the PPG signal in the time domain) and physiological features (e.g. HR and HRV). We run a feature selection algorithm, namely PCA (Principal Component Analysis) [32], to select the most useful features. This reduces the number of features to five. We use these features to train our predictive model. The first three features are obtained from the frequency domain. They are the amplitude of the highest peak, its corresponding frequency, and the ratio of the highest peak over the total power. If $y(t)$ is the PPG signal in the frequency domain, the latter feature can be written as:

$$\omega = \frac{\max |y(t)|^2}{E[|y(t)|^2]}$$

where $\max |y(t)|^2$ is the maximum power and $E[|y(t)|^2]$ is the expected value of $|y(t)|^2$.

The fourth and fifth features are in the time domain and are the standard deviation and mean of the distance between troughs and peaks in the PPG signal.

We performed preliminary testing on several classifiers to select the ones that yield the highest precision, recall, and accuracy. For instance, we tested decision tree, Support Vector Machine (SVM) and K-Nearest Neighbor (KNN). In Section V, we present the results we obtained for the best three classifiers: KNN, Random Forest, and K-star. KNN and K-Star are instance-based learning algorithms. KNN stores instances of the training data to classify the new test data point. Each test data point is considered in the same class of the nearest instance to the known test data point. The k -neighbors nearest to the test instance are selected and then the average of their predicted values is assigned to the test instance. K-Star uses the entropic distance of instances to classify the test dataset based on the probability of transforming an instance into another by randomly choosing between all possible transformations. Random Forest is an ensemble learning method that utilizes multiple decision trees to train the dataset and returns the mean prediction or mode of the classes from each decision tree. The Random Forest method has several strengths that render it an attractive classification approach. It can estimate which features are important for classification (inner feature selection algorithm). It reduces the chance of producing a model that overfits the training data and there is no need to perform tree pruning [33], [34].

Furthermore, we propose to combine the Random Forest and K-star classifiers through a stacking scheme to further improve the accuracy and precision of the algorithm. Stacking mixes several classifiers to make a final prediction from those of the combined classifiers.

The input to the ML algorithm is the extracted features from each component and the output is a "Yes/No" label of whether the component carries PPG information. Hence, the HR value calculated from the independent component labeled as "Yes" by the ML algorithm is the one adopted as the correct HR. Although unlikely, there is a possibility that we obtain two or three "Yes" classifications

from the three evaluated independent components. Therefore, the algorithm calculates a final HR estimate by averaging all HR values obtained from all components that correspond to a “Yes” classification. Similarly, if we obtain three “No” classifications, then we average the HR values obtained from all three components. We provide details about the dataset we use to train and test this algorithm in Section IV. In addition, we present the training procedure in Section IV-C.

E. HR ADJUSTMENT THROUGH LINEAR REGRESSION

The HR calculated using the frequency domain based MPA approach is shifted with respect to the time domain based HR estimate [35]. In fact, in our preliminary investigation, we observed that the estimated HR derived through the MPA approach exhibits a linear shift with respect to the HR extracted from the ECG sensor through an R peaks count. We use linear regression to model this phenomenon. A linear regression model finds a linear relationship between one dependent variable and other explanatory or independent variables. We use this model to further correct the HR we extract from the video PPG signal using the MPA approach.

We create a dataset that includes the HR obtained from ECG (as a dependent variable) and the video PPG based HR (as an explanatory variable). The dataset is described in Section IV.

Given our dataset $\{y_i, x_i\}_{i=1}^n$ of n data points, the model takes the form

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (2)$$

where $i = 1, \dots, n$ and β_0 and β_1 are model parameters. β_1 is the slope and β_0 is the intercept of the linear model. The relationship is modeled in the presence of the error variable ε . We assume $E(\varepsilon_i) = 0$ for all i . We use the Ordinary Least Squares (OLS) method, which minimizes the sum of squared residuals, to estimate β . Hence, we need to solve:

$$\min_{\beta_0, \beta_1} \sum (y_i - \beta_0 - \beta_1 x_i)^2 \quad (3)$$

The solution is given by:

$$\hat{\beta}_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} \quad (4)$$

$$\hat{\beta}_0 = \sum (y_i - \beta_1 x_i) \quad (5)$$

The goodness of the fit is assessed in Section V by the R^2 measure, and when there is only one explanatory variable, is given by:

$$R^2 = \frac{\text{Cov}(x, y)^2}{\text{Var}(x) \text{Var}(y)} \quad (6)$$

R^2 ranges between 0 and 1, and a higher measure indicates that the model possesses more explanatory power.

IV. DATASET

We collected a dataset to train and/or test the light equalization, ML, and linear regression methods proposed in

Sections III-B, III-D, and III-E respectively. The dataset collection procedure was approved by the Office of Research Ethics and Integrity at the University of Ottawa (Certificate Number: H02-17-13). Eleven adult subjects (4 female and 7 male) volunteered to participate and signed a consent form. We collected 220 video segments of the 11 subjects where each video segment is 1 minute long. The dataset consists of two subsets: Stationary and Movement Mode. The Stationary Mode subset is used in the experiment described in Section V-A. The Movement Mode subset is employed in the experiment detailed in Section V-B.

A. STATIONARY MODE SUBSET

We asked participants to sit in front of a camera for 5 minutes with minimum movement. They wore a physiological sensor (Zephyr Bioharness)¹ that collects an ECG signal at a sampling rate of 250 HZ. The data collected by the sensor is used as ground truth to evaluate the accuracy of our results. Fig. 3a shows the experimental setup for the stationary mode. The videos are collected by a conventional RGB camera (Logitech Webcam C270) at 30-frames per second and 640×480 pixels per frame—while the camera is mounted on a stable tripod at a one-meter distance from the subject. The lighting in the room is a mixture of natural sunlight coming from windows and fluorescent ceiling light.

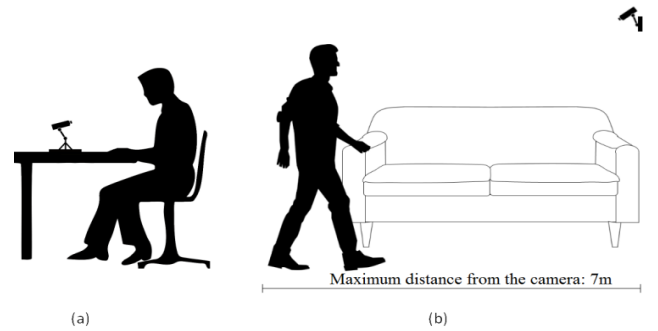


FIGURE 3. Experimental setup for (a) Stationary mode, (b) Movement mode.

B. MOVEMENT MODE SUBSET

As opposed to the Stationary Mode, for the Movement Mode subset, we asked the subjects to move in the room for 15 minutes by walking, talking, or lying down given that at least 50% of their face is visible to the camera (see Fig. 3b). The dimensions of the room are $7\text{m} \times 7\text{m}$. The rest of the experimental conditions, including lighting, camera, and sensor are identical to the ones used in the experiment described in Section IV-A.

Table 1 describes the conditions for the Stationary and Movement Mode subset collection.

¹“Zephyr TM Performance Systems | Performance Monitoring Technology.” [Online]. Available: <https://www.zephyranywhere.com/>. [Accessed: 17-Apr-2018].

TABLE 1. Description of the stationary and movement mode data subsets.

Parameters	Movement Mode	Stationary Mode
Experiment length per subject (second)	900	300
# of participants	11	11
	(4 F, 7 M)	(4 F, 7 M)
Distance from camera (meter)	0-7	1
Recording rate (fps)	30	30
# of 1 min. video segments	165	55
# of frames	27000	9000

C. TRAINING THE ML ALGORITHM

We train the ML algorithm we employ for channel selection post-ICA (proposed in Section III-D) using a randomly chosen 70% of the dataset (i.e. 154 video segments). We reserve 30% of the dataset for later testing. Each video is processed using the proposed method described in Section III and depicted in Fig. 1b up to the Filtering stage. The ICA module produces three independent components. Hence, from the 154 video segments, we extract 462 independent components (3 from each video). We manually attach a label of “Yes” or “No” to each independent component based on the ground truth HR as measured by the ECG sensor. Evidently, the “Yes” label corresponds to the independent component that best reflects the HR compared to the two other ones produced from the same video segment. That is, the “Yes” labeled independent component is the most appropriate to estimate the HR from the facial PPG. To establish the label for an independent component, we calculate the absolute difference between the HR estimated from that component and HR obtained from the ECG sensor. We suppose that the independent component that renders the smallest absolute difference is the one that best reflects the HR, and consequently is the one that is granted the “Yes” label. The other two are given the “No” label.

During training, the dataset is randomly shuffled to maintain generalized learning. By shuffling the data, we ensure that each data point creates an independent change on the model, without being biased by previous ones. We use a 10-fold cross-validation to assess the fitness of the model before evaluating it with the testing data set. Table 2 presents the cross-validation results of the ML models.

TABLE 2. Performance of the cross-validation of the proposed ML Technique.

Performance Metrics	KNN	Random Forest	K-Star	Stacking C (Random Forest, K-Star)
Precision	0.876	0.902	0.892	0.925
Recall	0.871	0.911	0.886	0.917
Accuracy (%)	85.896	91.291	87.452	92.135
Area under ROC Curve (AUC)	0.889	0.932	0.882	0.965

D. GENERATING THE LINEAR REGRESSION MODEL

To generate the linear regression model described in Section III-E, we randomly choose 70% of the dataset and calculate the HR using the MPA scheme. We reserve the remaining 30% for predictive assessment (see Section V). Then applying Equations (3), (4), and (5), we resolve the values of β_0 and β_1 which describe the model.

V. EVALUATION

In this section, we provide an evaluation of the proposed contributions as detailed in Section I. The dataset used in the experiments is described in Sections IV-A and IV-B.

A. HR ESTIMATION IN STATIONARY MODE

We present the results of the HR estimation from facial video in the Stationary Mode. Hence, we use the Stationary Mode data subset described in Section IV-A for all assessments.

1) ICA INDEPENDENT COMPONENT CHANNEL SELECTION

The ICA module outputs three channels, one of which contains the most accurate representation of the PPG signal. Fig. 4 shows an example of the independent components post-ICA and filtering. We propose an ML-based algorithm to identify the independent component that best reflects the PPG information.

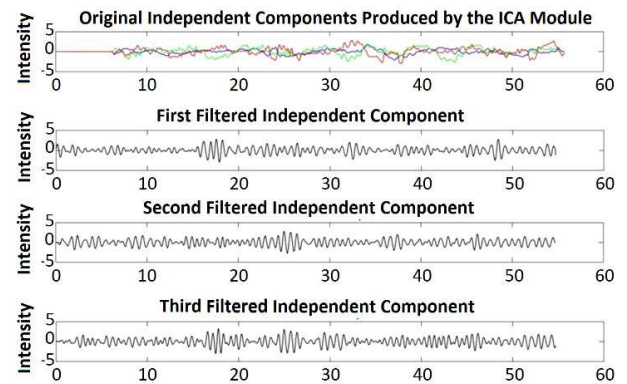
**FIGURE 4.** Output of the ICA and noise reduction procedures.

Table 3 presents the classification results for the four considered ML schemes: KNN, Random Forest, K-Star and Stacked Random Forest and K-Star. The Stacked Random Forest and K-Star yields the best results in terms of Precision, Recall, and Accuracy. This is unsurprising given that stacking two or more classifiers with reasonable performance typically increases the overall accuracy of the model [34]. Hence, we will adopt this scheme as our proposed ML-based method.

To select the independent component carrying the PPG signal, Poh *et al.* [3] heuristically choose the second component. Monkaresi *et al.* [5] propose an ML technique to choose the best independent component produced by the ICA module. They select nine features to train their ML algorithm. The features are the frequency with the highest magnitude in each

TABLE 3. Performance of the proposed ML Technique.

Performance Metrics	KNN	Random Forest	K-Star	Stacking C (Random Forest, K-Star)
Precision	0.848	0.854	0.852	0.870
Recall	0.845	0.855	0.849	0.869
Accuracy (%)	84.539	85.191	85.213	86.871
Area under ROC Curve (AUC)	0.869	0.912	0.871	0.934

component before and after noise filtering and the index of the spectral peak in the Power Spectral Density (PSD) of each component, after the application of the noise reduction method. Their algorithm utilizes ICA+KNN to select the best signal out of the three independent components. They find that the third component produces the lowest RMSE compared to the others. However, they did not report the performance of their ML scheme. They conclude that the third component is the most probable signal that carries the HR information.

We calculate the absolute difference between the HR obtained from ECG and the HR obtained from each of the three independent components outputted by the ICA module for a video segment. We count the independent component that results in the lowest absolute difference to be the one carrying the useful HR information. Through this process, we found that for our dataset, the HR is best calculated as follows by the three channels: 13.11% by the first component, 30.58% by the second component, and 56.31% by the third component. Although it seems from these results that the PPG signal is mostly carried by the second and third components, we cannot select a single channel that consistently best carries the PPG information. Table 4 compares the proposed ML-based method (Stacked Random Forest and K-Star) with that of Poh *et al.* [3] and Monkaresi *et al.* [5]. The proposed scheme produces better results compared to the approaches by Poh *et al.* [3] and Monkaresi *et al.* [5] in terms of Precision, Recall, and Accuracy. The ICA produces the independent components in a random order [36]. Hence, it might be counterproductive to assume that one component, based on

TABLE 4. Comparison of Poh *et al.* [3], Monkaresi *et al.* [5], and the proposed ML scheme for channel selection.

Performance Metrics	Poh <i>et al.</i> [3]	Monkaresi <i>et al.</i> [5]	Stacking C (Random Forest, K-Star)
Precision	0.271	0.463	0.870
Recall	0.266	0.455	0.869
Accuracy (%)	26.658	46.284	86.871
Area under ROC Curve (AUC)	51.356	60.472	0.934

its order as it appears in the output of the ICA module, will always best reflect the PPG signal. Hence, the application of the proposed ML algorithm to evaluate all independent components and resolve the best one removes any reliance on the order in which the independent components appear in the output. In the proposed ML scheme, the application of the PCA technique to narrow our initial feature set allowed us to preserve the most relevant features to solve the problem. Furthermore, the application of stacking permitted us to leverage the strengths of the Random Forest and K-Star algorithms.

2) HR ESTIMATION BEFORE LIGHT EQUALIZATION AND REGRESSION

Table 5 shows the comparison of the Poh *et al.* [3] and Monkaresi *et al.* [5] HR estimation techniques with the proposed method (without the application of light equalization and regression). We can observe the positive effect of accurately selecting the independent component through the proposed ML-based technique on the results. For instance, the RMSE decreased by 37% and 21% compared to Poh *et al.* [3] and Monkaresi *et al.* [5], respectively.

TABLE 5. Comparison of Poh *et al.* [3], Monkaresi *et al.* [5], and proposed method before light equalization in stationary mode.

Parameters	Poh <i>et al.</i> [3]	Monkaresi <i>et al.</i> [5]	Proposed method
Selected component	2 nd	3 rd	-
Mean Bias (bpm)	-0.12	0.91	0.79
SD of Bias (bpm)	3.11	2.03	1.34
Upper limit (bpm)	6.09	3.97	3.41
Lower limit (bpm)	-6.33	-2.15	-1.83
RMSE (bpm)	2.38	1.89	1.50
Corr. coefficient	0.98*	0.98*	0.98*

*: (p<0.004)

3) HR ESTIMATION AFTER LIGHT EQUALIZATION AND BEFORE REGRESSION

Table 6 shows the improvement in HR estimation when light equalization is applied. We achieved a lower RMSE in comparison with Poh *et al.* [3] and Monkaresi *et al.* [5]. By utilizing light equalization, the fluctuation of light across frames in a video segment is stabilized. These fluctuations can possibly introduce noise to the signal as they alter the skin color from which the PPG signal is derived. Hence, their reduction results in a cleaner signal as it is evidenced by the results of Table 5.

4) HR ESTIMATION AFTER LIGHT EQUALIZATION AND REGRESSION

The proposed regression model in Section III-E finds a relationship between the HR calculated from the video derived PPG signal and HR extracted from the ECG sensor. Using this

TABLE 6. Comparison of Poh *et al.* [3], Monkaresi *et al.* [5], and proposed method after light equalization in stationary mode.

Parameters	Poh <i>et al.</i> [3]	Monkaresi <i>et al.</i> [5]	Proposed method
Selected components	2 nd	3 rd	-
Mean Bias (bpm)	-0.12	0.91	0.78
SD of Bias (bpm)	3.11	2.03	1.32
Upper limit (bpm)	6.09	3.97	2.58
Lower limit (bpm)	-6.33	-2.15	-1.03
RMSE (bpm)	2.38	1.89	1.47
Corr. coefficient	0.98*	0.98*	0.98*

*: (p<0.004)

model, we are able to eliminate any systematic bias between the two HR extraction methods. Hence, when we applied the regression model, we obtained a lower mean and standard deviation of bias and a lower RMSE. Table 7 summarizes the details of the results before and after applying the regression model.

TABLE 7. Comparison of the estimated HR after light equalization and before and after regression in stationary mode.

Parameters	Before Regression	After Regression
Mean Bias (bpm)	0.78	0.70
SD of Bias (bpm)	1.32	1.25
Upper limit (bpm)	2.58	3.15
Lower limit (bpm)	-1.03	-1.75
RMSE	1.47	1.38
Corr. coefficient	0.98*	0.99*

*: (p<0.004)

B. HR ESTIMATION IN MOVEMENT MODE

Table 8 compares the estimated HR with subjects' movements before and after the light equalization. As in the Stationary Mode, we observe a decrease in the RMSE when light equalization is applied.

TABLE 8. Comparison of the estimated hr before and after light equalization in movement mode.

Parameters	Before Light Equalization	After Light Equalization
Mean Bias (bpm)	2.12	1.47
SD of Bias (bpm)	3.96	3.10
Upper limit (bpm)	7.76	5.78
Lower limit (bpm)	-3.48	-3.46
RMSE	2.43	1.69
Corr. coefficient	0.88*	0.89*

*: (p<0.004)

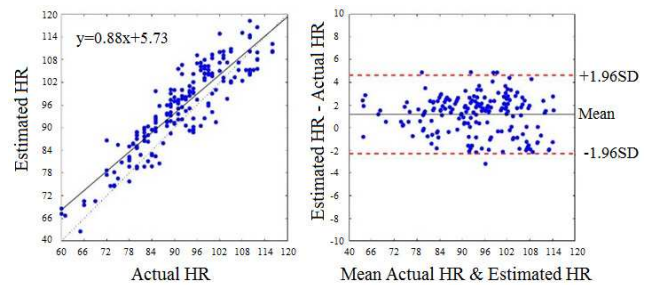
Table 9 shows the results of the proposed algorithm when light equalization and regression are applied. We achieved a 34% reduction in RMSE compared to the experiment when the regression was not applied.

TABLE 9. Comparison of the estimated HR after light equalization and before and after regression in movement mode.

Parameters	Before Regression	After Regression
Mean Bias (bpm)	1.47	1.20
SD of Bias (bpm)	3.10	1.77
Upper limit (bpm)	5.78	4.70
Lower limit (bpm)	-3.46	-2.30
RMSE	1.69	1.12
Corr. coefficient	0.89*	0.93*

*: (p<0.004)

Fig. 5 reports the results of the regression model and the Bland Altman plot on the Actual HR and our estimated HR. The Bland Altman plot shows the high agreement between the actual HR and the estimated HR.

**FIGURE 5.** Statistical Results of (a) Regression model (b) Bland and Altman.

VI. DISCUSSION

One of the properties of ICA is that the source signals are returned in a random order [36]. Therefore, making an assumption about which component contains the most useful information based on the order of the output of the ICA module contradicts the nature of ICA. Monkaresi *et al.* [5] use ICA and an ML model to predict the best component. They indicate that the output of their algorithm chooses the third component as the PPG carrying signal among all others; hence, this is the channel they adopt. Poh *et al.* [3] heuristically choose the second component as the PPG reflecting the signal. However, in the proposed method, the signal selection runs every time a video is processed to extract the HR. No assumption is made about the order in which the independent components are outputted by the ICA module. The independent component carrying the PPG information is dynamically calculated. The results of Table 4 show the advantage of this dynamic approach over the static schemes of Monkaresi *et al.* [5] and Poh *et al.* [3] in terms of improving the Precision, Recall, and Accuracy of the independent component selection.

Furthermore, for the channel selection scheme, we employ a stacked ML algorithm to optimize performance. We combine a group of “weaker learners” to realize a “stronger learner”. We tested several ML Classifiers and the best results are achieved by the stacked Random forest and K-star (Table 3). Random Forest uses a bagging approach to increase the accuracy of a classifier. Random forest combines several decision trees to improve the overall performance of the model. The K-star algorithm uses entropy as a distance measure for classification. It provides a consistent approach to handle the real-valued attributes and missing values. In fact, Cleary and Trigg [37] compared several instance-based algorithms in classification. They utilize several popular datasets commonly used in ML literature. They conclude that K-star performs well across all dataset and has superiority over all instance-based algorithms.

Although the light equalization technique may have improved the performance by lowering the RMSE by 2% (Tables 5 and 6) in the Stationary Mode, we noticed a larger effect in the Movement Mode where we observed an RMSE decrease of 30% (Table 8). In comparison to the Stationary Mode, the positive effect of light equalization is larger in the presence of movement as the light projected on the face fluctuates more in this scenario. As the subject displaces in the unevenly lit room, the amount and direction of light reflecting on the face vary, thus emphasizing the need for light equalization to reduce the negative effect of such phenomenon on the quality of the retrieved PPG signal.

We assess a linear relationship between HR calculated from the video derived PPG signal and HR extracted from the ECG sensor. We calculate R^2 , a statistical measure that describes the goodness of the fit, to evaluate the regression performance. R^2 shows how closely the data fit the regression model. The range of R^2 is between 0 and 1. The value of 0 indicates that the regression model does not explain the variation in the predicted variable around its mean, and 1 indicates that the model explains impeccably the variation in the predicted variable around its mean. Fig.5 shows the high agreement of the estimated HR with the actual HR extracted from the sensor ($R^2 = 0.88$). Given a model slope smaller than 1 and an intercept larger than 0, and an R^2 value indicating a high explanatory power for the model, we presume that a shift between the HR obtained from the ECG sensor and HR obtained through the MPA approach from facial PPG does exist and can be linearly modeled. Such a model allows us to improve the HR estimation as evidenced by the results of Table 7 (Stationary Mode) and Table 9 (Movement Mode).

VII. CONCLUSION

We proposed and evaluated a method for the remote measurement of HR using a PPG signal extracted from a video recording of a subject's face. The proposed method is based on the technique presented by Poh et al.'s [3]. However, we extended the latter technique by implementing a light equalization scheme to reduce the effect of spatial and temporal light

variation on the HR estimation, an ML method to select the most accurate channel outputted by the ICA module, and a regression model that estimates the relationship between the HR calculated through the MPA scheme and the measurement extracted from an ECG signal.

In our experimental results, we found that the proposed light equalization technique reduces the RMSE by 0.74 bpm for moving subjects (Table 8) and 0.03 bpm for stationary subjects (Tables 5 and 6), the ML technique selects the correct component outputted by ICA with an accuracy of 86.9%, (Table 3) and the application of the regression model to adjust the estimated HR reduced the RMSE by 0.57 bpm for moving subjects (Table 9) and 0.09 bpm for stationary subjects (Table 7).

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